Lab08:Unsupervised Learning Mini-Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyse a biosbsy data-set from fine needle aspiration (FNA) of a breast mass.

Data Import

The dara is made available as a CSV file for download. We can read this using `read.csv()`"

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
```

```
head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

Make sure we remove or exclude the `diagnosis` column from the data-st that we use for further analysis- this is the expert diagnosis as either M or B.

```r
# We can us -1 to remove the first column

wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data

Q1. How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

There are 212 malignant diagnosis

```r
# It will show all the variables number of the specific column u ask
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

OR

```r
sum(wisc.df$diagnosis=="M")
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with _mean?

```r
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"
 [3] "perimeter_mean"          "area_mean"
 [5] "smoothness_mean"         "compactness_mean"
 [7] "concavity_mean"          "concave.points_mean"
 [9] "symmetry_mean"           "fractal_dimension_mean"
[11] "radius_se"               "texture_se"
[13] "perimeter_se"            "area_se"
[15] "smoothness_se"           "compactness_se"
[17] "concavity_se"            "concave.points_se"
[19] "symmetry_se"             "fractal_dimension_se"
[21] "radius_worst"            "texture_worst"
[23] "perimeter_worst"         "area_worst"
[25] "smoothness_worst"        "compactness_worst"
[27] "concavity_worst"         "concave.points_worst"
[29] "symmetry_worst"          "fractal_dimension_worst"
```

We can use the `grep()` function to help us here to find how many column have the mean suffix in it.

```r
## the length shows how many the grep shows where are they in columns

length(grep(pattern = "_mean", colnames(wisc.data) ))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```r
wisc.pr <- prcomp(wisc.data, scale. = T)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
```

```
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> **Q4.** From your results, what proportion of the original variance is captured by the first principal component (PC1)?

44.27% of the vairance is captured by PC1

> **Q5.** How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

To reach at least 70% of the original variance total of first PC 3 is required.

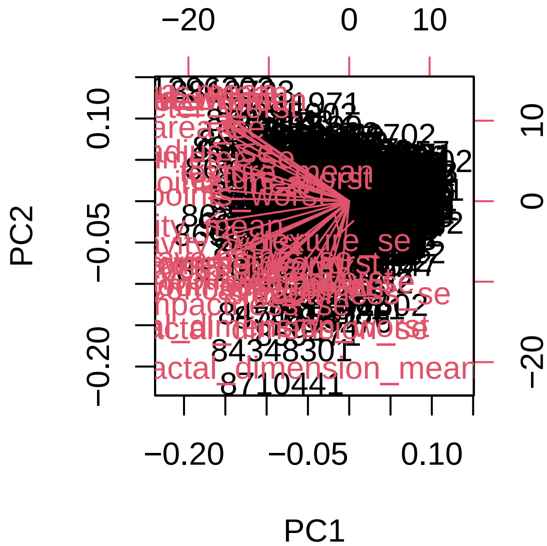
> **Q6.** How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

To reach at least 90% of the original variance, 7 principal components are required, since the cumulative variance at PC7 is about 91%.

Interpreting PCA Results

```r
## The biplot creates a plot of all datas from dataset.

biplot(wisc.pr)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

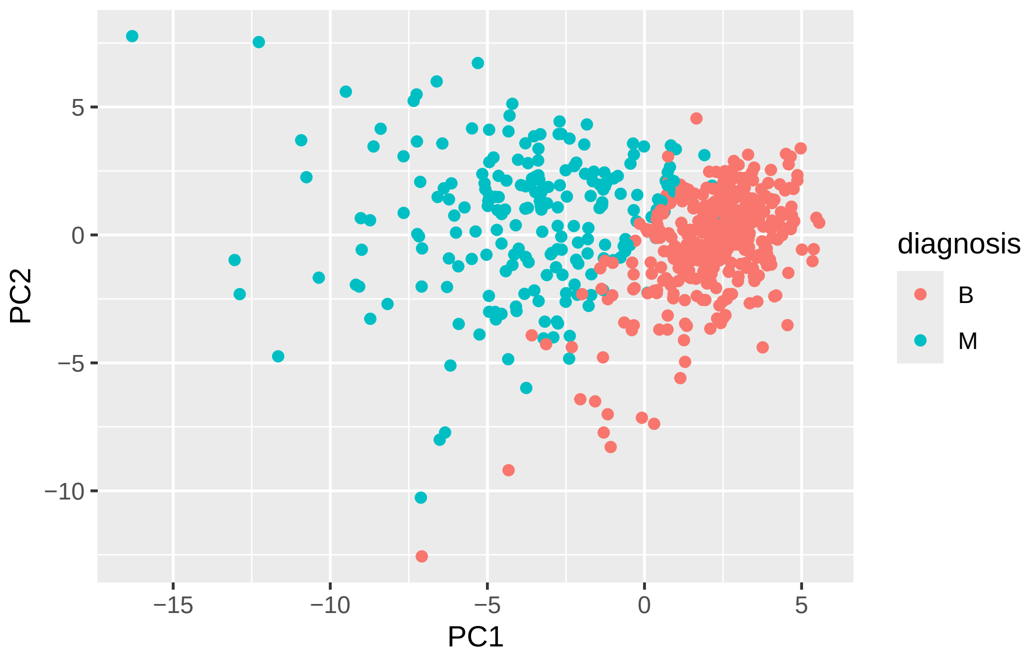
The plot is very messy and the result it show is very weak and almost impossible to read, with lots of overlapping points and labels, making it hard to clearly see patterns. It is difficult to understand because the data points are all clustered together in the center and the variable arrows overlap, so nothing is clearly separated. Even though the biplot contains a lot of information, it's not very readable in this form.

Let's see the "PC Score Plot" for the first 2 component of the data which are PC1 and PC2.

```r
# Scatter plot observations by components 1 and 2

library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```

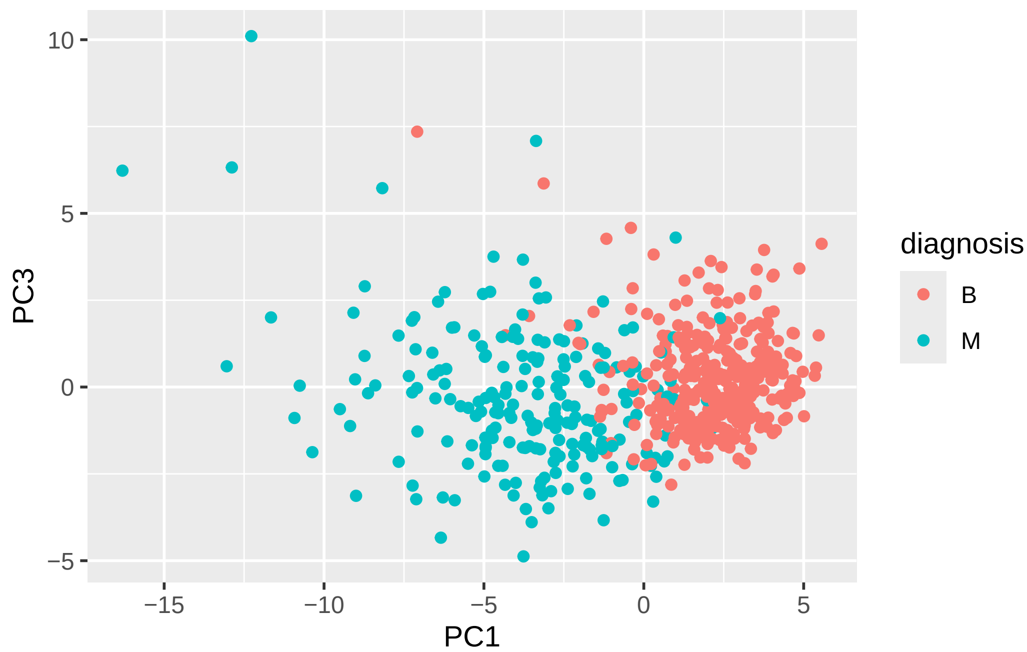


Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
# Scatter plot observations by components 1 and 3

library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



Scree Plot

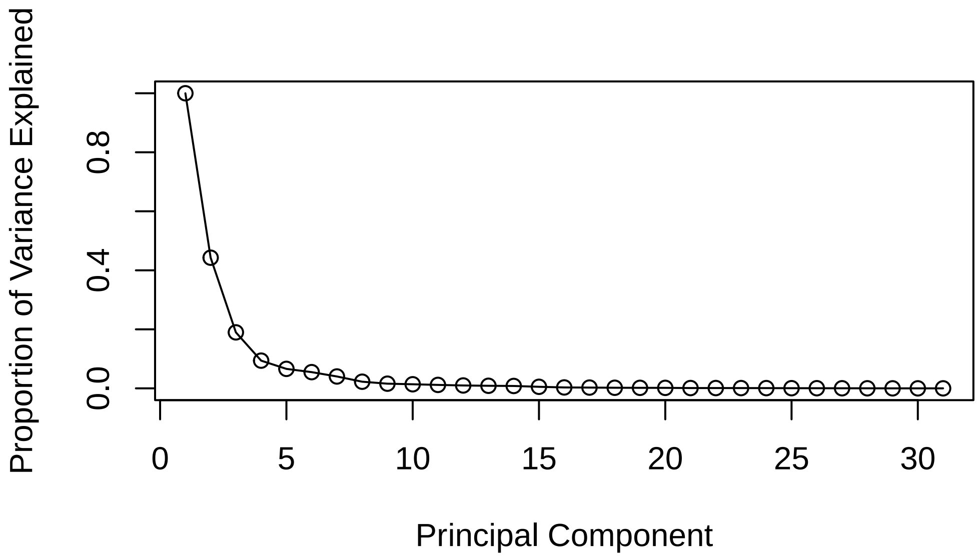
Calculate the variance of each principal component by squaring the sdev component of wisc.pr (i.e. wisc.pr$sdev^2). Save the result as an object called pr.var.

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

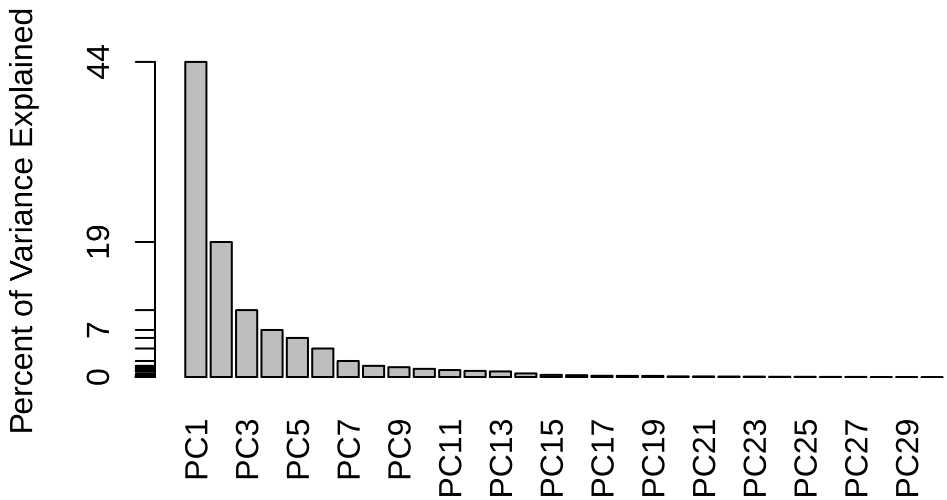
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```r
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", "PC1"]
```

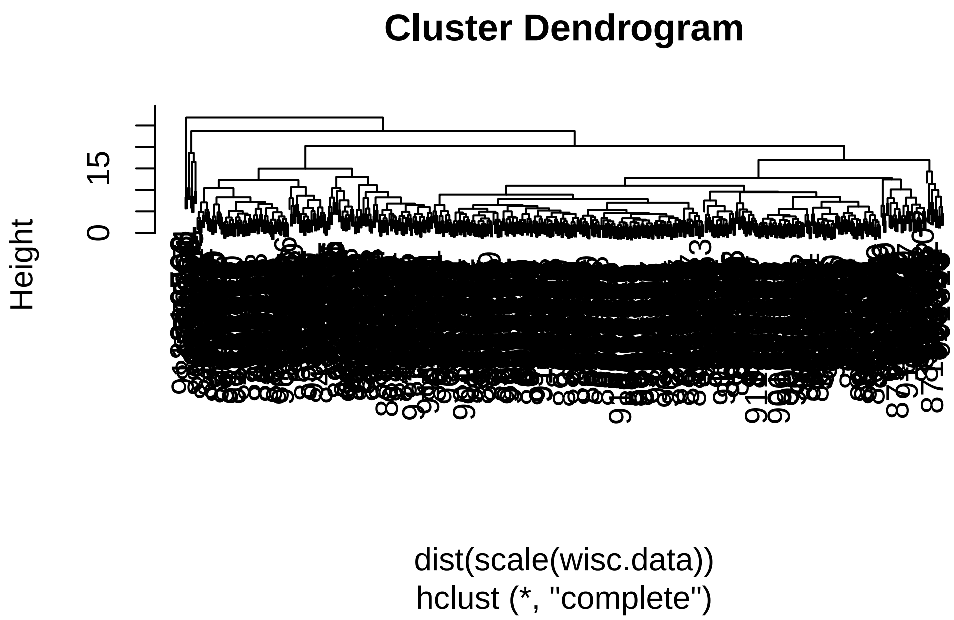
```
[1] -0.2608538
```

The loading value for concave.points_mean in PC1 is about -0.26, which means it has a fairly strong contribution to the first principal component. The negative sign just indicates the direction.

Heirarchical Clustering

Now from scree plot we can choose which PCs contain most of the vairance and that is PC1-PC4.

```
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```

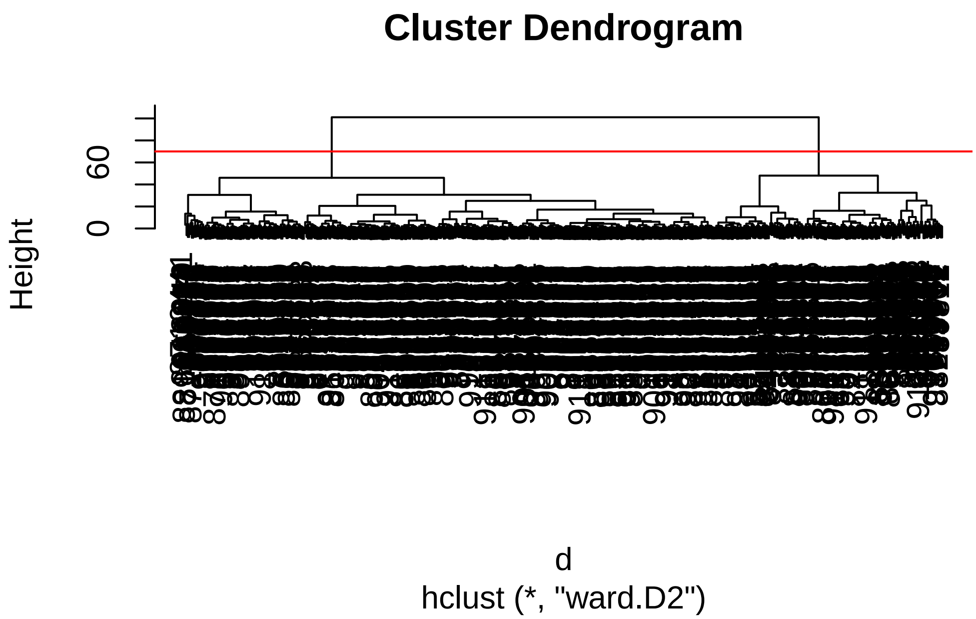


> **Q10.** Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

The hieght of 4 clusters combined is approximately 70.

Combining Method

```
d <- dist(wisc.pr$x[, 1:4])
wisc.pr.hclust <- hclust(d, method = "ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```



Selecting Number of Clusters

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
grps <- cutree(wisc.pr.hclust, h=70, k=2)
table(grps)
```

```
grps
  1   2
171 398
```

How does clustering `grps` correspond to the expert diagnosis?

```
table(diagnosis)
```

```
diagnosis
  B   M 
357 212 
```

```
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

> **Q11.** OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

```
grps2 <- cutree(wisc.pr.hclust, k=2)
table(diagnosis, grps2)
```

```
         grps2
diagnosis   1   2
        B   6 351
        M 165  47
```

```
grps3 <- cutree(wisc.pr.hclust, k=3)
table(diagnosis, grps3)
```

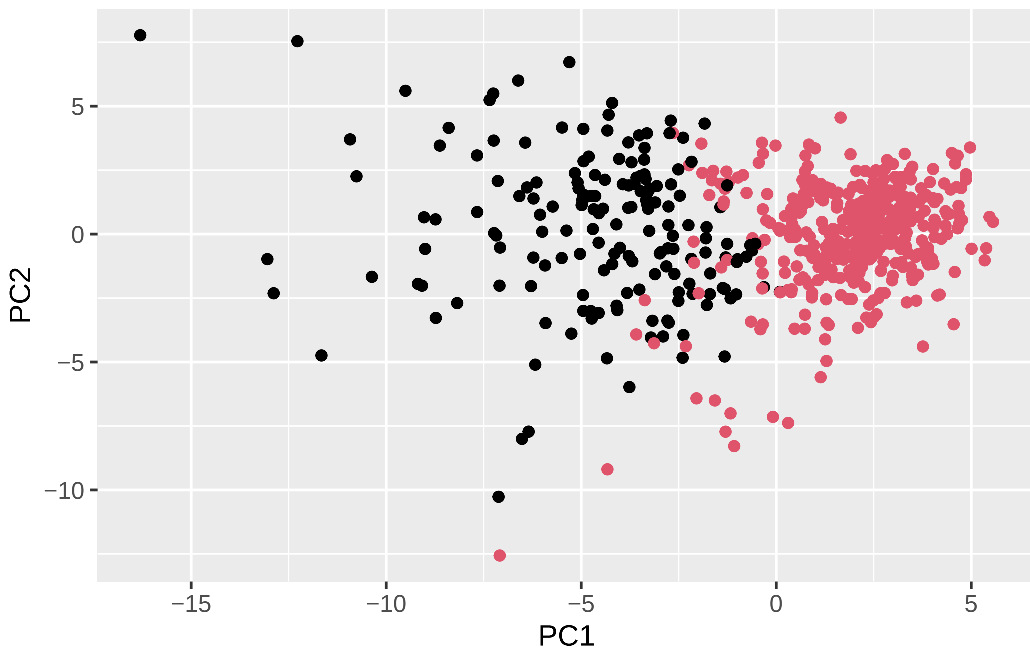
```
         grps3
diagnosis   1   2   3
        B   0   6 351
        M 104  61  47
```

```
grps4 <- cutree(wisc.pr.hclust, k=4)
table(diagnosis, grps4)
```

```
          grps4
diagnosis   1   2   3   4
        B   0   6 267  84
        M 104  61  41   6
```

Clustering with 2 groups already does a good job separating benign and malignant samples, with most benign cases grouped together and many malignant cases in another cluster. Trying different numbers of clusters (like 3 or 4) have slightly improved separation, but can also made the results harder to interpret.

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



> **Q12.** Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

The ward.D2 method gives the best results for this dataset. It creates clusters that are more balanced and compact, with clearer separation between groups compared to the other methods.

```r
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method="ward.D2")
```

```r
# Step 2: cut into 2 clusters
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)

# Step 3: compare with diagnosis
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

> **Q13.** How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

The new clustering using the first 7 PCs separates benign and malignant samples slightly better, with clearer grouping and less noise. Most benign samples fall into one cluster and most malignant samples into another, although a few misclassifications still remain.

> **Q14.**How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```r
# With PCA (first 7 PCs)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

The hierarchical clustering without PCA separates the diagnoses fairly well, with one cluster mainly containing benign samples and another mainly containing malignant samples. However, there are some smaller clusters that contain mixed diagnoses, showing that the separation is not perfect.

Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

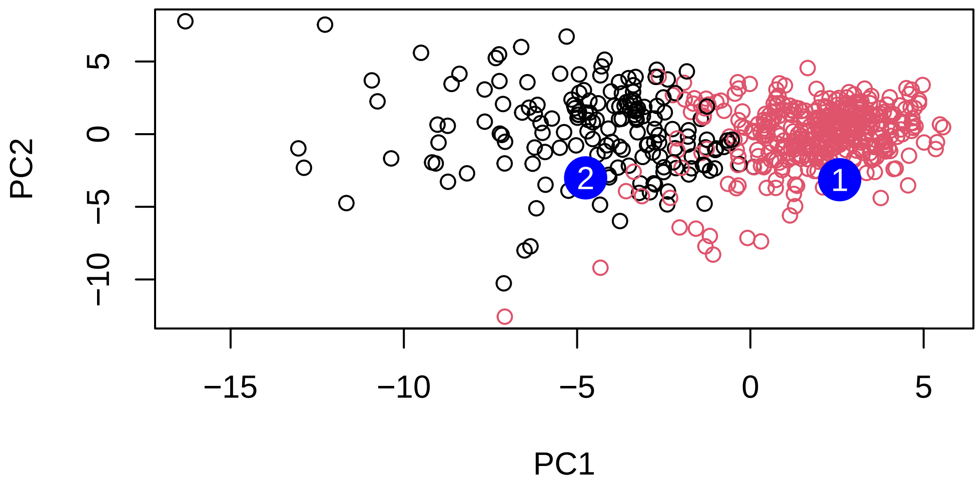
The clustering based on the first 7 principal components showed the best performance overall. It had higher sensitivity because it correctly identified more malignant cases, and also maintained good specificity by grouping most benign cases together. In comparison, the clustering without PCA had more mixing between groups, leading to slightly lower sensitivity and specificity.

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2 is closer to the cluster that contains more malignant (black points), while patient 1 is clearly grouped with the benign cluster (red points). This suggests that patient 2 has features more similar to malignant cases and may be at higher risk, so, patient 2 should be checked first.